First-class topology carriers

The explicit type value retains its classifier annotation. The named carrier and both identity functions are defined in Clean. These three statements have separately checked Lean proofs in the isolated prototype.

Result 1 (explicit topology carrier). [Local, identity]
Let X be a topological space. Then

$$\text{retain in}((\text{Top} : \text{Types}_1), X) = X.$$

Result 2 (inferred topology carrier). [Local, identity]
Let X be a topological space. Then

$$\text{retain inferred}(X) = X.$$

Result 3 (named topology carrier). [Local, identity]
Let X be a topological space. Then

$$\text{retain in}(\text{topology carrier}(), X) = X.$$

LLM-assisted source and proofs (gpt-6, 2026-11-11).